Main Routine

Cycle Ring Buffer
v3.02 Dec2024 - replace handshake bit with TransNum (complete) or negative TransNum (resend)
integer TransNum eliminates the Read Complete Delay Time since the Gateway does not need to reset the handshake bit
v3.50 - L980 Controller

0

S:FS

Program Version

aoiEDC_ProgVersion	
aoiEDC_ProgVersion	EDC_aoiProgVersion
Major	3
Minor	5
Build	0
Patch	0
Version	Prog.Version

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE	
Source	1101
Dest	Prog.Routine 1120

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD	
Source A	Prog.RoutineScan[Prog.Routine] -806558467
Source B	1
Dest	Prog.RoutineScan[Prog.Routine] -806558467

Confgure and Initialize Ring Buffer Pointers and Transacations
v3.02e - add ConfigONS

2

S:FS

EDC Config
Configuration ONS
(Log Transaction)
CFG.ConfigONS
<EDC.CFG.ConfigONS>

Configuration

JSR	
Routine Name	_1111_Config

Main Routine

Process RingBuffer Reads by Transaction Manager to SQL

3

Process RingBuffer Reads from Transaction Manager

JSR
Routine Name _1120_RingBuffer

Capture Last Scan Time for this Progam
(Common Routine in all Progams)

(this routine must be after Controller Status for Program Scan total to be correct)

4

CLx Module Status Clear Scan History
CLX.ProgScan.Clear
<EDC.CLX.ProgScan.Clear>

Local Program Scan Time & Fault Routine Clear Scan History
Prog.Scan.Clear
L

Local Program Scan Time & Fault Routine Program Fault Occurrences

CLR
Dest Prog.FaultCount
0

Program Scan

aoiEDC_ProgScan
aoiEDC_ProgScan EDC_aoiProgScan ...
Scan Prog.Scan
ScanTemp CLX.ProgScanTemp
<EDC.CLX.ProgScanTemp>

Shift Reset
Store Prev Shift and Clear Current Shift

5

ShiftReset

EDC Ring Buffer Control Diagnostic - Auto Update every shift

MOVE
Source RB.ShiftCnt
<EDC.RB_Cycle01.ShiftCnt>
7564
Dest RB.PrevShiftCnt
<EDC.RB_Cycle01.PrevShiftCnt>
8603

EDC Ring Buffer Control Diagnostic - Auto Reset every shift

CLR
Dest RB.ShiftCnt
<EDC.RB_Cycle01.ShiftCnt>
7564

Main Routine

Program Scan Exceeds High Limit (configured for 100 msec)

6

Local Program Scan Time & Fault Routine Last Scan

GT
Source A: Prog.Scan.Last 13
Source B: CLxScanHighLimit 100000

Program High Scan Count

ADD
Source A: ProgScanHighCount 0
Source B: 1
Dest: ProgScanHighCount 0

Program High Scan Events

CLxProgScanHigh.11 (L)

CLx Module Status Clear Scan History

CLX.ProgScan.Clear
<EDC.CLX.ProgScan.Clear>

Program High Scan Count

CLR
Dest: ProgScanHighCount 0

(End)

Fault Routine

Program Fault Routine
(Common Routine in all Progams)

This routine is invoked by CLx Major Faults. It logs the Fault, clears the Fault and attempts to resume program scan.

0

NOP

Get Fault Record and Increment Fault Count

1

Local Program Scan Time & Fault Routine Low 32 bits of fault timestamp value.

GSV

Class Name	Program
Instance Name	THIS
Attribute Name	MAJORFAULTRECORD
Dest	Prog.Fault.TimeLow
	0

Local Program Scan Time & Fault Routine Program Fault Occurrences

ADD

Source A	Prog.FaultCount
	0
Source B	1
Dest	Prog.FaultCount
	0

Fault Routine

Increment Log Fault Index -- Determine Index limits based on Log Array Size
(this allows Array to be resized without logic changes)

2

Local Program Scan Time & Fault Routine Fault Routine Log

SIZE	
Source	Prog.Log[0] ??
Dim. To Vary	0
Size	LEN 1000

Local Program Scan Time & Fault Routine Program Fault Log Maximum Index

SUB	
Source A	LEN 1000
Source B	1
Dest	Prog.LOGmi 0

Local Program Scan Time & Fault Routine Program Fault Log Index

ADD	
Source A	Prog.LOGi 0
Source B	1
Dest	Prog.LOGi 0

Local Program Scan Time & Fault Routine Program Fault Log Index

LT	
Source A	Prog.LOGi 0
Source B	0

Local Program Scan Time & Fault Routine Program Fault Log Index

GT	
Source A	Prog.LOGi 0
Source B	Prog.LOGmi 0

Local Program Scan Time & Fault Routine Program Fault Log Index

CLR	
Dest	Prog.LOGi 0

Fault Routine

Log Fault in Local Program Log
Optional Parameters are for the Indexes used in this Program -- the likely cause of the Major Faults.

3

Local Program Scan Time & Fault Routine Fault Record
COP
Source Prog.Fault
Dest Prog.Log[Prog.LOGi].Fault
Length 1

Local Program Scan Time & Fault Routine Fault Counter
MOVE
Source Prog.FaultCount
0
Dest Prog.Log[Prog.LOGi].Count
0

Local Program Scan Time & Fault Routine Faulted Routine Number
MOVE
Source Prog.Routine
1120
Dest Prog.Log[Prog.LOGi].Routine
0

Local Program Scan Time & Fault Routine Transaction Scan Couner
MOVE
Source CLX.ScanCount
<EDC.CLX.ScanCount>
466525176
Dest Prog.Log[Prog.LOGi].ScanCount
0

Local Program Scan Time & Fault Routine Transaction Date (YYMMDD)
MOVE
Source CLX.SysTime[8]
<EDC.CLX.SysTime[8]>
20261030
Dest Prog.Log[Prog.LOGi].Date
0

Local Program Scan Time & Fault Routine Transaction Time (HHMMSS)
MOVE
Source CLX.SysTime[9]
<EDC.CLX.SysTime[9]>
151921
Dest Prog.Log[Prog.LOGi].Time
0

Local Program Scan Time & Fault Routine Transaction Time (microseconds)
MOVE
Source CLX.SysTime[11]
<EDC.CLX.SysTime[11]>
904
Dest Prog.Log[Prog.LOGi].TimeuSec
0

Local Program Scan Time & Fault Routine Optional Parameter 1 (defined by Transaction)
MOVE
Source RB.WriteTagPtr
<EDC.RB_Cycle01.WriteTagPtr>
2
Dest Prog.Log[Prog.LOGi].Opt1
0

Local Program Scan Time & Fault Routine Optional Parameter 2 (defined by Transaction)
MOVE
Source RB.WriteRecPtr
<EDC.RB_Cycle01.WriteRecPtr>
547
Dest Prog.Log[Prog.LOGi].Opt2
0

Local Program Scan Time & Fault Routine Optional Parameter 3 (defined by Transaction)
MOVE
Source RB.ReadTagPtr
<EDC.RB_Cycle01.ReadTagPtr>
2
Dest Prog.Log[Prog.LOGi].Opt3
0

Local Program Scan Time & Fault Routine Optional Parameter 4 (defined by Transaction)
MOVE
Source RB.ReadRecPtr
<EDC.RB_Cycle01.ReadRecPtr>
546
Dest Prog.Log[Prog.LOGi].Opt4
0

Local Program Scan Time & Fault Routine Optional Parameter 5 (defined by Transaction)
MOVE
Source 0
Dest Prog.Log[Prog.LOGi].Opt5
0

Log Fault Routine Event to Transaction Log
Unable to call TransactionLogSBR from Fault Routine. Main Routine logs the Fault after Program Execution has resumed.
(Only one Fault Routine event can be logged per scan -- but LOGi and Local Fault Log will indicate multiple Events.)

4

Local Program Scan Time & Fault Routine Fault Routine Executed -- Log in Transaction Log
Prog.LogFault
L

Fault Routine

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

5

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 1102
Dest Prog.Routine 1120

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[Prog.Routine] -806558467
Source B 1
Dest Prog.RoutineScan[Prog.Routine] -806558467

Clear Fault to Resume Program Scan
(Disable Fault Handler to allow Major Controller Faults to occur for debug.)

6

EDC Config Disable CLx Processor Fault Handler (for Testing Only)
CFG.DisableFaultHandler
<EDC.CFG.DisableFaultHandler>

Local Program Scan Time & Fault Routine Program Clear Fault Record

COP
Source Prog.Fault
Dest Prog.ClearFault
Length 1

Local Program Scan Time & Fault Routine Fault Type (Program, I/O, etc.)

MOVE
Source 0
Dest Prog.ClearFault.Type 0

Local Program Scan Time & Fault Routine Unique (by Type) Code for Fault.

MOVE
Source 0
Dest Prog.ClearFault.Code 0

Local Program Scan Time & Fault Routine Low 32 bits of fault timestamp value.

SSV
Class Name Program
Instance Name THIS
Attribute Name MAJORFAULTRECORD
Source Prog.ClearFault.TimeLow 0

(End)

Configuration

Ring Buffer Compare

0

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 1111
Dest Prog.Routine 1120

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[Prog.Routine] -806558467
Source B 1
Dest Prog.RoutineScan[Prog.Routine] -806558467

Configure Ring Buffer Array size and Tag Count
(Ring Buffers with multiple tags MUST use same array size on all Tags.)

2

Equipment Data Cycle Message

SIZE
Source EDC_RB_Cycle01_Rec01[0] ??
Dim. To Vary 0
Size LEN 1000

EDC Ring Buffer Control Maximum Record Pointer

SUB
Source A LEN 1000
Source B 1
Dest RB.MaxRecPtr <EDC.RB_Cycle01.MaxRecPtr> 999

EDC Ring Buffer Control Maximum Tag Pointer

MOVE
Source 5
Dest RB.MaxTagPtr <EDC.RB_Cycle01.MaxTagPtr> 5

Configuration

Check for uninitialized Transaction Number

3

EDC Ring Buffer Control TM Transaction Trigger

EQ
Source A RB.TransactionNumber <EDC.RB_Cycle01.TransactionNumber> 6814
Source B 0

EDC Ring Buffer Control TM Transaction Trigger

MOVE
Source 1
Dest RB.TransactionNumber <EDC.RB_Cycle01.TransactionNumber> 6814

Reset Ring Buffer Faults caused by missing Config

4

EDC Ring Buffer Control Diagnostic - Manual Reset RB.WritePtrErr <EDC.RB_Cycle01.WritePtrErr>
U

EDC Ring Buffer Control Diagnostic - Manual Reset RB.ReadPtrErr <EDC.RB_Cycle01.ReadPtrErr>
U

(End)

Process RingBuffer Reads from Transaction Manager

Extract Read Records from RingBuffer for Transaction Manager
(supports multiple Tags for Ring Buffers with >2M data)

NOTE: Configure Tag Count in _1111_Config Rung 2

0

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 1120
Dest Prog.Routine 1120

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[Prog.Routine] -806558467
Source B 1
Dest Prog.RoutineScan[Prog.Routine] -806558467

Process RingBuffer Reads from Transaction Manager

Reset Pointers if Read Pointer outside limits
(this should never happen - but protects the Array Indexes)

2

EDC Ring Buffer Control Current/Last Read Tag

LT
Source A RB.ReadTagPtr <EDC.RB_Cycle01.ReadTagPtr> 2
Source B 1

EDC Ring Buffer Control Current/Last Read Tag

GT
Source A RB.ReadTagPtr <EDC.RB_Cycle01.ReadTagPtr> 2
Source B RB.MaxTagPtr <EDC.RB_Cycle01.MaxTagPtr> 5

EDC Ring Buffer Control Current/Last Read Record

LT
Source A RB.ReadRecPtr <EDC.RB_Cycle01.ReadRecPtr> 546
Source B 0

EDC Ring Buffer Control Current/Last Read Record

GT
Source A RB.ReadRecPtr <EDC.RB_Cycle01.ReadRecPtr> 546
Source B RB.MaxRecPtr <EDC.RB_Cycle01.MaxRecPtr> 999

Manually Reset Error Indicator & Tags

EDC Ring Buffer Control Diagnostic - Manual Reset

RB.ReadPtrErr <EDC.RB_Cycle01.ReadPtrErr>

(L)

EDC Ring Buffer Control Diagnostic - Manual Reset Value

MOVE
Source RB.ReadTagPtr <EDC.RB_Cycle01.ReadTagPtr> 2
Dest RB.ReadPtrErrTag <EDC.RB_Cycle01.ReadPtrErrTag> 0

EDC Ring Buffer Control Diagnostic - Manual Reset Value

MOVE
Source RB.ReadRecPtr <EDC.RB_Cycle01.ReadRecPtr> 546
Dest RB.ReadPtrErrRec <EDC.RB_Cycle01.ReadPtrErrRec> 0

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB.WriteTagPtr <EDC.RB_Cycle01.WriteTagPtr> 2
Dest Trans.Opt[1] <EDC.Trans.Opt[1]> -1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB.WriteRecPtr <EDC.RB_Cycle01.WriteRecPtr> 547
Dest Trans.Opt[2] <EDC.Trans.Opt[2]> -1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB.ReadTagPtr <EDC.RB_Cycle01.ReadTagPtr> 2
Dest Trans.Opt[3] <EDC.Trans.Opt[3]> -1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB.ReadRecPtr <EDC.RB_Cycle01.ReadRecPtr> 546
Dest Trans.Opt[4] <EDC.Trans.Opt[4]> -1

Log Transactions

JSR
Routine Name _901_TransLogSBR
Input Par 2523
Input Par 0
Input Par -1
Input Par -1
Input Par RB.TransactionNumber <EDC.RB_Cycle01.TransactionNumber>

EDC Ring Buffer Control Next Write Tag

MOVE
Source 1
Dest RB.WriteTagPtr <EDC.RB_Cycle01.WriteTagPtr> 2

EDC Ring Buffer Control Next Write Record

MOVE
Source 0
Dest RB.WriteRecPtr <EDC.RB_Cycle01.WriteRecPtr> 547

EDC Ring Buffer Control Current/Last Read Tag

MOVE
Source RB.MaxTagPtr <EDC.RB_Cycle01.MaxTagPtr> 5
Dest RB.ReadTagPtr <EDC.RB_Cycle01.ReadTagPtr> 2

EDC Ring Buffer Control Current/Last Read Record

MOVE
Source RB.MaxRecPtr <EDC.RB_Cycle01.MaxRecPtr> 999
Dest RB.ReadRecPtr <EDC.RB_Cycle01.ReadRecPtr> 546

Process RingBuffer Reads from Transaction Manager

Next Read Location
ReadTagPointer & ReadPointer = Current/Last Read Record (Incremented before Read)
ReadNextTagPointer & ReadNextPointer = Next Read Record (if/when Written)

3

EDC Ring Buffer Control Next Read Record

GT
Source A RB.ReadNextRecPtr <EDC.RB_Cycle01.ReadNextRecPtr> 547
Source B RB.MaxRecPtr <EDC.RB_Cycle01.MaxRecPtr> 999

EDC Ring Buffer Control Next Read Tag

MOVE
Source RB.ReadTagPtr <EDC.RB_Cycle01.ReadTagPtr> 2
Dest RB.ReadNextTagPtr <EDC.RB_Cycle01.ReadNextTagPtr> 2

EDC Ring Buffer Control Next Read Record

ADD
Source A RB.ReadRecPtr <EDC.RB_Cycle01.ReadRecPtr> 546
Source B 1
Dest RB.ReadNextRecPtr <EDC.RB_Cycle01.ReadNextRecPtr> 547

EDC Ring Buffer Control Next Read Tag

ADD
Source A RB.ReadNextTagPtr <EDC.RB_Cycle01.ReadNextTagPtr> 2
Source B 1
Dest RB.ReadNextTagPtr <EDC.RB_Cycle01.ReadNextTagPtr> 2

EDC Ring Buffer Control Next Read Tag

GT
Source A RB.ReadNextTagPtr <EDC.RB_Cycle01.ReadNextTagPtr> 2
Source B RB.MaxTagPtr <EDC.RB_Cycle01.MaxTagPtr> 5

EDC Ring Buffer Control Next Read Record

MOVE
Source 0
Dest RB.ReadNextRecPtr <EDC.RB_Cycle01.ReadNextRecPtr> 547

EDC Ring Buffer Control Next Read Tag

MOVE
Source 1
Dest RB.ReadNextTagPtr <EDC.RB_Cycle01.ReadNextTagPtr> 2

Process RingBuffer Reads from Transaction Manager

Buffer Full - Read Pointer = Write Pointer
WritePointer = Next Write Record (Incremented after Write)

Buffer Empty -- Read Pointer next to Write Pointer
no more messages in Ring Buffer
(RB Empty when Read complete - don't increment Read Pointer)

4

EDC Ring Buffer Control Current/Last Read Tag

EQ
Source A RB.ReadTagPtr <EDC.RB_Cycle01.ReadTagPtr> 2
Source B RB.WriteTagPtr <EDC.RB_Cycle01.WriteTagPtr> 2

EDC Ring Buffer Control Current/Last Read Record

EQ
Source A RB.ReadRecPtr <EDC.RB_Cycle01.ReadRecPtr> 546
Source B RB.WriteRecPtr <EDC.RB_Cycle01.WriteRecPtr> 547

EDC Ring Buffer Control Buffer Full
RB.Full
<EDC.RB_Cycle01.Full>

EDC Ring Buffer Control Read Buffer Record in Progress
RB.ReadInProgress
<EDC.RB_Cycle01.ReadInProgress>

EDC Ring Buffer Control Next Read Tag

EQ
Source A RB.ReadNextTagPtr <EDC.RB_Cycle01.ReadNextTagPtr> 2
Source B RB.WriteTagPtr <EDC.RB_Cycle01.WriteTagPtr> 2

EDC Ring Buffer Control Next Read Record

EQ
Source A RB.ReadNextRecPtr <EDC.RB_Cycle01.ReadNextRecPtr> 547
Source B RB.WriteRecPtr <EDC.RB_Cycle01.WriteRecPtr> 547

EDC Ring Buffer Control Buffer Empty
RB.Empty
<EDC.RB_Cycle01.Empty>

5

Buffer Pointers
Combine Tag & Record Pointers

EDC Ring Buffer Control Next Write Tag/Record

CPT
Dest RB.WritePtr <EDC.RB_Cycle01.WritePtr> 1547
Expression (RB.WriteTagPtr-1)*(RB.MaxRecPtr+1)+RB.WriteRecPtr

EDC Ring Buffer Control Current/Last Read Tag/Record

CPT
Dest RB.ReadPtr <EDC.RB_Cycle01.ReadPtr> 1546
Expression (RB.ReadTagPtr-1)*(RB.MaxRecPtr+1)+RB.ReadRecPtr

EDC Ring Buffer Control Next Read Tag/Record

CPT
Dest RB.ReadNextPtr <EDC.RB_Cycle01.ReadNextPtr> 1547
Expression (RB.ReadNextTagPtr-1)*(RB.MaxRecPtr+1)+RB.ReadNextRecPtr

Process RingBuffer Reads from Transaction Manager

Buffer Count

6

EDC Ring Buffer Control Read Buffer Record in Progress RB.ReadInProgress <EDC.RB_Cycle01.ReadInProgress>

EDC Ring Buffer Control Next Write Tag/Record

GT
Source A RB.WritePtr <EDC.RB_Cycle01.WritePtr> 1547
Source B RB.ReadPtr <EDC.RB_Cycle01.ReadPtr> 1546

EDC Ring Buffer Control Ring Buffer Read Record Count

CPT
Dest RB.RecordCount <EDC.RB_Cycle01.RecordCount> 0
Expression RB.WritePtr-RB.ReadPtr

EDC Ring Buffer Control Next Write Tag/Record

LE
Source A RB.WritePtr <EDC.RB_Cycle01.WritePtr> 1547
Source B RB.ReadPtr <EDC.RB_Cycle01.ReadPtr> 1546

EDC Ring Buffer Control Ring Buffer Read Record Count

CPT
Dest RB.RecordCount <EDC.RB_Cycle01.RecordCount> 0
Expression (RB.MaxRecPtr+1)*RB.MaxTagPtr-(RB.ReadPtr-RB.WritePtr)

EDC Ring Buffer Control Read Buffer Record in Progress RB.ReadInProgress <EDC.RB_Cycle01.ReadInProgress>

EDC Ring Buffer Control Next Write Tag/Record

GE
Source A RB.WritePtr <EDC.RB_Cycle01.WritePtr> 1547
Source B RB.ReadNextPtr <EDC.RB_Cycle01.ReadNextPtr> 1547

EDC Ring Buffer Control Ring Buffer Read Record Count

CPT
Dest RB.RecordCount <EDC.RB_Cycle01.RecordCount> 0
Expression RB.WritePtr-RB.ReadNextPtr

EDC Ring Buffer Control Next Write Tag/Record

LT
Source A RB.WritePtr <EDC.RB_Cycle01.WritePtr> 1547
Source B RB.ReadNextPtr <EDC.RB_Cycle01.ReadNextPtr> 1547

EDC Ring Buffer Control Ring Buffer Read Record Count

CPT
Dest RB.RecordCount <EDC.RB_Cycle01.RecordCount> 0
Expression (RB.MaxRecPtr+1)*RB.MaxTagPtr-(RB.ReadNextPtr-RB.WritePtr)

EDC Ring Buffer Control Ring Buffer Read Record Count

GT
Source A RB.RecordCount <EDC.RB_Cycle01.RecordCount> 0
Source B RB.RecordCountMax <EDC.RB_Cycle01.RecordCountMax>

EDC Ring Buffer Control Ring Buffer Read Record Maximum Count

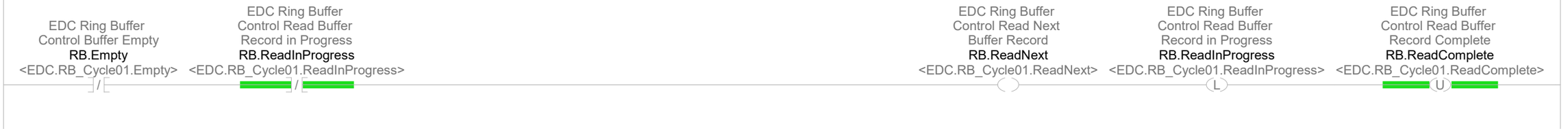
MOVE
Source RB.RecordCount <EDC.RB_Cycle01.RecordCount> 0
Dest RB.RecordCountMax <EDC.RB_Cycle01.RecordCountMax>

Process RingBuffer Reads from Transaction Manager

3 3

Read Next
Ready for next Transaction Manager Read
DHF019 - Remove ONS - not needed and allowed ReadNext to hang on power up

7



Process RingBuffer Reads from Transaction Manager

Read Next
Increment Read Pointer

8

EDC Ring Buffer Control Read Next Buffer Record
RB.ReadNext
<EDC.RB_Cycle01.ReadNext>

EDC Ring Buffer Control Current/Last Read Record
ADD
Source A RB.ReadRecPtr <EDC.RB_Cycle01.ReadRecPtr> 546
Source B 1
Dest RB.ReadRecPtr <EDC.RB_Cycle01.ReadRecPtr> 546

EDC Ring Buffer Control Current/Last Read Record
LT
Source A RB.ReadRecPtr <EDC.RB_Cycle01.ReadRecPtr> 546
Source B 0

EDC Ring Buffer Control Current/Last Read Record
GT
Source A RB.ReadRecPtr <EDC.RB_Cycle01.ReadRecPtr> 546
Source B RB.MaxRecPtr <EDC.RB_Cycle01.MaxRecPtr> 999

EDC Ring Buffer Control Current/Last Read Record
MOVE
Source 0
Dest RB.ReadRecPtr <EDC.RB_Cycle01.ReadRecPtr> 546

EDC Ring Buffer Control Current/Last Read Tag
ADD
Source A RB.ReadTagPtr <EDC.RB_Cycle01.ReadTagPtr> 2
Source B 1
Dest RB.ReadTagPtr <EDC.RB_Cycle01.ReadTagPtr> 2

EDC Ring Buffer Control Current/Last Read Tag
LT
Source A RB.ReadTagPtr <EDC.RB_Cycle01.ReadTagPtr> 2
Source B 1

EDC Ring Buffer Control Current/Last Read Tag
GT
Source A RB.ReadTagPtr <EDC.RB_Cycle01.ReadTagPtr> 2
Source B RB.MaxTagPtr <EDC.RB_Cycle01.MaxTagPtr> 5

EDC Ring Buffer Control Current/Last Read Tag
MOVE
Source 1
Dest RB.ReadTagPtr <EDC.RB_Cycle01.ReadTagPtr> 2

Process RingBuffer Reads from Transaction Manager

Process RingBuffer Reads from Transaction Manager

Read Next
Load Read Record (Tags 01-05)

9

EDC Ring Buffer
Control Read Next
Buffer Record
RB.ReadNext
<EDC.RB_Cycle01.ReadNext>

EDC Ring Buffer Control Current/Last Read Tag	Equipment Data Cycle Message
EQ Source A: RB.ReadTagPtr <EDC.RB_Cycle01.ReadTagPtr> Source B: 2 1	COP Source: EDC_RB_Cycle01_Rec01[RB.ReadRecPtr] Dest: RB_Read <EDC_RB_Cycle01_Read> Length: 1
EQ Source A: RB.ReadTagPtr <EDC.RB_Cycle01.ReadTagPtr> Source B: 2 2	COP Source: EDC_RB_Cycle01_Rec02[RB.ReadRecPtr] Dest: RB_Read <EDC_RB_Cycle01_Read> Length: 1
EQ Source A: RB.ReadTagPtr <EDC.RB_Cycle01.ReadTagPtr> Source B: 2 3	COP Source: EDC_RB_Cycle01_Rec03[RB.ReadRecPtr] Dest: RB_Read <EDC_RB_Cycle01_Read> Length: 1
EQ Source A: RB.ReadTagPtr <EDC.RB_Cycle01.ReadTagPtr> Source B: 2 4	COP Source: EDC_RB_Cycle01_Rec04[RB.ReadRecPtr] Dest: RB_Read <EDC_RB_Cycle01_Read> Length: 1
EQ Source A: RB.ReadTagPtr <EDC.RB_Cycle01.ReadTagPtr> Source B: 2 5	COP Source: EDC_RB_Cycle01_Rec05[RB.ReadRecPtr] Dest: RB_Read <EDC_RB_Cycle01_Read> Length: 1

Process RingBuffer Reads from Transaction Manager

Read Next
Increment Transaction Number
(Triggers TM Transaction on change)

10

EDC Ring Buffer
Control Read Next
Buffer Record
RB.ReadNext
<EDC.RB_Cycle01.ReadNext>

EDC Ring Buffer
Control Read Buffer
Record in Progress
Timeout
RB.ReadInProgressTimeout
<EDC.RB_Cycle01.ReadInProgressTimeout>

EDC Ring Buffer
Control TM
Transaction Trigger

ADD
Source A RB.TransactionNumber
<EDC.RB_Cycle01.TransactionNumber>
6814
Source B 1
Dest RB.TransactionNumber
<EDC.RB_Cycle01.TransactionNumber>
6814

EDC Ring Buffer
Control TM
Transaction Trigger

LT
Source A RB.TransactionNumber
<EDC.RB_Cycle01.TransactionNumber>
6814
Source B 1

EDC Ring Buffer
Control TM
Transaction Trigger

GT
Source A RB.TransactionNumber
<EDC.RB_Cycle01.TransactionNumber>
6814
Source B CFG.MsgTransNumRollover
<EDC.CFG.MsgTransNumRollover>
9999

EDC Ring Buffer
Control TM
Transaction Trigger

MOVE
Source 1
Dest RB.TransactionNumber
<EDC.RB_Cycle01.TransactionNumber>
6814

Process RingBuffer Reads from Transaction Manager

Read Next
Calculate LNi, EQi for Transaction Logs & Performance

11

EDC Ring Buffer Control Read Next Buffer Record
RB.ReadNext
<EDC.RB_Cycle01.ReadNext>

EDC Ring Buffer Control Read Buffer Record in Progress Timeout
RB.ReadInProgressTimeout
<EDC.RB_Cycle01.ReadInProgressTimeout>

MOD
Source A: RB_Read.Header.EQID <EDC_RB_Cycle01_Read.Header.EQID> 965
Source B: 100
Dest: EQi 19

CPT
Dest: LNi 4
Expression: (RB_Read.Header.EQID-EQi)/100

CLR
Dest: PerfLNi 4

CLR
Dest: PerfEQi 19

EDC Ring Buffer Control Read Data has Valid Equipment ID
RB.ReadValidEQID
<EDC.RB_Cycle01.ReadValidEQID>
(U)

EDC Ring Buffer Control Read Data has Valid Equipment ID
RB.ReadValidEQID
<EDC.RB_Cycle01.ReadValidEQID>
(L)

LIMIT
Low Limit: 0
Test: LNi 4
High Limit: CFG.LN.mi <EDC.CFG.LN.mi> 28

LIMIT
Low Limit: 0
Test: EQi 19
High Limit: CFG.EQ.mi <EDC.CFG.EQ.mi> 99

MOVE
Source: LNi 4
Dest: PerfLNi 4

MOVE
Source: EQi 19
Dest: PerfEQi 19

Copy Equipment Arrayed Data to Working Tags
Working Tags are used for speed and logic readabliity (ie. can be xref'd)
DN007 Split EquipStatus Tag

12

EDC Ring Buffer Control Read Data has Valid Equipment ID
RB.ReadValidEQID
<EDC.RB_Cycle01.ReadValidEQID>

Local Copy of Equip Array Tag for LNi, EQi

COP
Source: EDC_Status01.Line[PerfLNi].Equip[PerfEQi]
Dest: EquipStatus01
Length: 1

Equipment Data Collection - Status Line Equipment

COP
Source: EDC_Status04.Line[PerfLNi].Equip[PerfEQi]
Dest: EquipStatus04
Length: 1

Process RingBuffer Reads from Transaction Manager

13

Read Next
Log Transaction & Performance

EDC Ring Buffer
Control Read Next
Buffer Record
RB.ReadNext
<EDC.RB_Cycle01.ReadNext>

EDC Ring Buffer
Control Read Buffer
Record in Progress
Timeout
RB.ReadInProgressTimeout
<EDC.RB_Cycle01.ReadInProgressTimeout>

Transaction Log
Control SBR External
Load -- Optional
Parameters

MOVE
Source RB.WriteTagPtr
<EDC.RB_Cycle01.WriteTagPtr>
2
Dest Trans.Opt[1]
<EDC.Trans.Opt[1]>
-1

Transaction Log
Control SBR External
Load -- Optional
Parameters

MOVE
Source RB.WriteRecPtr
<EDC.RB_Cycle01.WriteRecPtr>
547
Dest Trans.Opt[2]
<EDC.Trans.Opt[2]>
-1

Transaction Log
Control SBR External
Load -- Optional
Parameters

MOVE
Source RB.ReadTagPtr
<EDC.RB_Cycle01.ReadTagPtr>
2
Dest Trans.Opt[3]
<EDC.Trans.Opt[3]>
-1

Transaction Log
Control SBR External
Load -- Optional
Parameters

MOVE
Source RB.ReadRecPtr
<EDC.RB_Cycle01.ReadRecPtr>
546
Dest Trans.Opt[4]
<EDC.Trans.Opt[4]>
-1

Log Transactions

JSR
Routine Name _901_TransLogSBR
Input Par 2520
Input Par 0
Input Par LNi
Input Par EQi
Input Par RB.TransactionNumber
<EDC.RB_Cycle01.TransactionNumber>

EDC Ring Buffer
Control Read Data
has Valid Equipment
ID
RB.ReadValidEQID
<EDC.RB_Cycle01.ReadValidEQID>

Equipment Data
Collection - Line
Equipment Control
for Message
Transaction Timer
Accum Last (used to
calculate
differential)

CLR
Dest TempPerfMsg.TransLast
110

Equipment Data
Collection - Line
Equipment Control
for Message
Transaction Timer

CLR
Dest TempPerfMsg.TransTMR.ACC
110

Performance
Transaction

aoiEDC_PerfTrans
aoiEDC_PerfTrans EDC_aoiPerfTrans ...
PerfMsg TempPerfMsg
PerfTrans EquipStatus04.CycleTx.EN

Process RingBuffer Reads from Transaction Manager

14

Read Complete
Wait for TM Handshake to Acknowledge Read Complete

EDC Ring Buffer
Control Read Buffer
Record in Progress
RB.ReadInProgress
<EDC.RB_Cycle01.ReadInProgress>

EDC Ring Buffer
Control TM
Transaction Trigger

EQ
Source A RB.TransactionNumber
<EDC.RB_Cycle01.TransactionNumber>
6814
Source B RB.Interface
<EDC.RB_Cycle01.Interface>
6814

EDC Ring Buffer
Control Read Buffer
Record Complete
RB.ReadComplete
<EDC.RB_Cycle01.ReadComplete>
L

EDC Ring Buffer
Control TM Handshake
negative value

NEG
Source RB.Interface
<EDC.RB_Cycle01.Interface>
6814
Dest RB.InterfaceNEG
<EDC.RB_Cycle01.InterfaceNEG>
-6814

EDC Ring Buffer
Control TM
Transaction Trigger

EQ
Source A RB.TransactionNumber
<EDC.RB_Cycle01.TransactionNumber>
6814
Source B RB.InterfaceNEG
<EDC.RB_Cycle01.InterfaceNEG>
-6814

EDC Ring Buffer
Control Read Buffer
Record Resend
RB.ReadResend
<EDC.RB_Cycle01.ReadResend>
L

Process RingBuffer Reads from Transaction Manager

Read Complete Delay
Don't reset TM Handshake too soon or it will cause a TM Error on Read Verify
No longer need delaly -- replaced Handshake bit with echo of TransNum
Slow/Fast delay added - no delay caused TM thread errors

15

EDC Ring Buffer Control Read Buffer Record Complete
RB.ReadComplete
<EDC.RB_Cycle01.ReadComplete>

Server Status TM Transaction OK Enable High Speed RB
ServerStat_TMokTMR.DN
/

EDC Ring Buffer Control Read Complete Delay Timer
MOVE
Source CfgRbDelaySlow 100
Dest RB.ReadCompleteDelayTMR.PRE <EDC.RB_Cycle01.ReadCompleteDelayTMR.PRE> 0

EDC Ring Buffer Control Read Complete Delay Timer
TON
Timer RB.ReadCompleteDelayTMR <EDC.RB_Cycle01.ReadCompleteDelayTMR>
Preset 0
Accum 13
EN
DN

Server Status TM Transaction OK Enable High Speed RB
ServerStat_TMokTMR.DN

EDC Ring Buffer Control Read Complete Delay Timer
MOVE
Source CfgRbDelayFast 0
Dest RB.ReadCompleteDelayTMR.PRE <EDC.RB_Cycle01.ReadCompleteDelayTMR.PRE> 0

EDC Ring Buffer Control Read Complete Delay Timer
RB.ReadCompleteDelayTMR.DN
<EDC.RB_Cycle01.ReadCompleteDelayTMR.DN>

EDC Ring Buffer Control Read Buffer Record Complete Delay
RB.ReadCompleteDelay
<EDC.RB_Cycle01.ReadCompleteDelay>

EDC Ring Buffer Control Read Buffer Record in Progress
RB.ReadInProgress
<EDC.RB_Cycle01.ReadInProgress>
U

Process RingBuffer Reads from Transaction Manager

16

EDC Ring Buffer Control Read Buffer Record Complete
RB.ReadComplete
<EDC.RB_Cycle01.ReadComplete>

EDC Ring Buffer Control Read Buffer Record Complete 1-shot
RB.ReadCompleteONS
<EDC.RB_Cycle01.ReadCompleteONS>
ONS

Read Complete
Log Transaction & Performance

EDC Ring Buffer Control Read Data has Valid Equipment ID
RB.ReadValidEQID
<EDC.RB_Cycle01.ReadValidEQID>

Local Copy of Equip Array Tag for LNi, EQi State = Green - Cycle Tx DN
EquipStatus01.Update.CycleTxOK
L

Update Status State for Transactions

aoiEDC_StatusUpdate

aoiEDC_StatusUpdate	aoiEDC_StatusUpdate ...
Status	EquipStatus01

Transaction Log Control SBR External Load -- Optional Parameters

MOVE

Source	RB.WriteTagPtr <EDC.RB_Cycle01.WriteTagPtr> 2
Dest	Trans.Opt[1] <EDC.Trans.Opt[1]> -1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE

Source	RB.WriteRecPtr <EDC.RB_Cycle01.WriteRecPtr> 547
Dest	Trans.Opt[2] <EDC.Trans.Opt[2]> -1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE

Source	RB.ReadTagPtr <EDC.RB_Cycle01.ReadTagPtr> 2
Dest	Trans.Opt[3] <EDC.Trans.Opt[3]> -1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE

Source	RB.ReadRecPtr <EDC.RB_Cycle01.ReadRecPtr> 546
Dest	Trans.Opt[4] <EDC.Trans.Opt[4]> -1

Log Transactions

JSR

Routine Name	_901_TransLogSBR
Input Par	2521
Input Par	0
Input Par	LNi
Input Par	EQi
Input Par	RB.TransactionNumber <EDC.RB_Cycle01.TransactionNumber>

EDC Ring Buffer Control Read Data has Valid Equipment ID
RB.ReadValidEQID
<EDC.RB_Cycle01.ReadValidEQID>

Equipment Data Collection - Line Equipment Control for Message Transaction Timer Accum Last (used to calculate differential)

CLR

Dest	TempPerfMsg.TransLast 110

Equipment Data Collection - Line Equipment Control for Message Transaction Timer

MOVE

Source	RB.ReadInProgressTMR.ACC <EDC.RB_Cycle01.ReadInProgressTMR.ACC> 0
Dest	TempPerfMsg.TransTMR.ACC 110

Performance Transaction

aoiEDC_PerfTrans

Process RingBuffer Reads from Transaction Manager

aoiEDC_PerfTrans EDC_aoiPerfTrans ...
PerfMsg TempPerfMsg
PerfTrans EquipStatus04.CycleTx.DN

Read in Progress - No Response from TM Timer
Increment Transaction Number to Retry TM Message Trigger

reduced retry from 62 to 5 seconds

17

EDC Ring Buffer Control Read Buffer Record in Progress RB.ReadInProgress <EDC.RB_Cycle01.ReadInProgress>

EDC Ring Buffer Control Read Buffer Record Complete RB.ReadComplete <EDC.RB_Cycle01.ReadComplete> /

EDC Ring Buffer Control Read Buffer Record Resend RB.ReadResend <EDC.RB_Cycle01.ReadResend> /

EDC Ring Buffer Control Read in Progress Timer

MOVE
Source 5000
Dest RB.ReadInProgressTMR.PRE <EDC.RB_Cycle01.ReadInProgressTMR.PRE> 5000

EDC Ring Buffer Control Read Buffer Record Resend RB.ReadResend <EDC.RB_Cycle01.ReadResend>

EDC Ring Buffer Control Read Buffer Record Resend RB.ReadResend <EDC.RB_Cycle01.ReadResend> U

EDC Ring Buffer Control Read in Progress Timer

MOVE
Source 2000
Dest RB.ReadInProgressTMR.PRE <EDC.RB_Cycle01.ReadInProgressTMR.PRE> 5000

EDC Ring Buffer Control Read in Progress Timer

TON
Timer RB.ReadInProgressTMR <EDC.RB_Cycle01.ReadInProgressTMR> EN
Preset 5000 DN
Accum 0

EDC Ring Buffer Control Read in Progress Timer RB.ReadInProgressTMR.DN <EDC.RB_Cycle01.ReadInProgressTMR.DN>

EDC Ring Buffer Control Read Buffer Record in Progress Timeout RB.ReadInProgressTimeout <EDC.RB_Cycle01.ReadInProgressTimeout>

Process RingBuffer Reads from Transaction Manager

Read in Progress - No Response from TM Timer
Log Transaction & Performance

18

EDC Ring Buffer Control Read Buffer Record in Progress Timeout
RB.ReadInProgressTimeout
<EDC.RB_Cycle01.ReadInProgressTimeout>

EDC Ring Buffer Control Read Data has Valid Equipment ID
RB.ReadValidEQID
<EDC.RB_Cycle01.ReadValidEQID>

Local Copy of Equip Array Tag for LNi, EQi State = Red - Cycle Tx Msg ER
EquipStatus01.Update.CycleTxER
L

Update Status State for Transactions
aoiEDC_StatusUpdate
aoiEDC_StatusUpdate aoiEDC_StatusUpdate ...
Status EquipStatus01

Transaction Log Control SBR External Load -- Optional Parameters
MOVE
Source RB.WriteTagPtr
<EDC.RB_Cycle01.WriteTagPtr>
2
Dest Trans.Opt[1]
<EDC.Trans.Opt[1]>
-1

Transaction Log Control SBR External Load -- Optional Parameters
MOVE
Source RB.WriteRecPtr
<EDC.RB_Cycle01.WriteRecPtr>
547
Dest Trans.Opt[2]
<EDC.Trans.Opt[2]>
-1

Transaction Log Control SBR External Load -- Optional Parameters
MOVE
Source RB.ReadTagPtr
<EDC.RB_Cycle01.ReadTagPtr>
2
Dest Trans.Opt[3]
<EDC.Trans.Opt[3]>
-1

Transaction Log Control SBR External Load -- Optional Parameters
MOVE
Source RB.ReadRecPtr
<EDC.RB_Cycle01.ReadRecPtr>
546
Dest Trans.Opt[4]
<EDC.Trans.Opt[4]>
-1

Log Transactions
JSR
Routine Name _901_TransLogSBR
Input Par 2524
Input Par 0
Input Par LNi
Input Par EQi
Input Par RB.TransactionNumber
<EDC.RB_Cycle01.TransactionNumber>

EDC Ring Buffer Control Read Data has Valid Equipment ID
RB.ReadValidEQID
<EDC.RB_Cycle01.ReadValidEQID>

Equipment Data Collection - Line Equipment Control for Message Transaction Timer Accum Last (used to calculate differential)
CLR
Dest TempPerfMsg.TransLast
110

Equipment Data Collection - Line Equipment Control for Message Transaction Timer
MOVE
Source RB.ReadInProgressTMR.ACC
<EDC.RB_Cycle01.ReadInProgressTMR.ACC>
0
Dest TempPerfMsg.TransTMR.ACC
110

Performance Transaction
aoiEDC_PerfTrans
aoiEDC_PerfTrans EDC_aoiPerfTrans ...
PerfMsg TempPerfMsg
PerfTrans EquipStatus04.CycleTx.ER

EDC Ring Buffer Control Read in Progress Timer
RB.ReadInProgressTMR
<EDC.RB_Cycle01.ReadInProgressTMR>
RES

Process RingBuffer Reads from Transaction Manager

RB Write -- Keep Alive Periodic Resend Last Transaction
Resend last message if RingBuffer is EMPTY for Keep Alive to TM/SQL
(Server will discard duplicate Transaction Numbers)

19

EDC Ring Buffer Control Read Buffer Record Complete Delay
RB.ReadCompleteDelay
<EDC.RB_Cycle01.ReadCompleteDelay>

EDC Ring Buffer Control Buffer Empty
RB.Empty
<EDC.RB_Cycle01.Empty>

EDC Ring Buffer Control Keep Alive Timer

MOVE
Source 300000
Dest RB.KeepAliveTMR.PRE
<EDC.RB_Cycle01.KeepAliveTMR.PRE>
300000

EDC Ring Buffer Control Keep Alive Timer

TON
Timer RB.KeepAliveTMR
<EDC.RB_Cycle01.KeepAliveTMR>
Preset 300000
Accum 34823
EN
DN

EDC Ring Buffer Control Keep Alive Timer
RB.KeepAliveTMR.DN
<EDC.RB_Cycle01.KeepAliveTMR.DN>

EDC Ring Buffer Control Read Buffer Keep Alive Resend Last
RB.KeepAlive
<EDC.RB_Cycle01.KeepAlive>

JSR
Routine Name _1121_RingBufferKeepAlive

EDC Ring Buffer Control Keep Alive Timer
RB.KeepAliveTMR
<EDC.RB_Cycle01.KeepAliveTMR>
RES

RingBuffer Alarm

Alarm if the RingBuffer is not Empty for 5 minutes.
(This will provide time to resolve the problem BEFORE the RingBuffer is full and data is lost.)

20

EDC Ring Buffer Control Buffer Empty
RB.Empty
<EDC.RB_Cycle01.Empty>
/

EDC Ring Buffer Control Buffer Not Empty Alarm Timer

MOVE
Source 300000
Dest RB.NotEmptyTMR.PRE
<EDC.RB_Cycle01.NotEmptyTMR.PRE>
300000

EDC Ring Buffer Control Buffer Not Empty Alarm Timer

TON
Timer RB.NotEmptyTMR
<EDC.RB_Cycle01.NotEmptyTMR>
Preset 300000
Accum 0
EN
DN

EDC Ring Buffer Control Buffer Not Empty Alarm Timer

GT
Source A RB.NotEmptyTMR.ACC
<EDC.RB_Cycle01.NotEmptyTMR.ACC>
0
Source B RB.NotEmptyMax
<EDC.RB_Cycle01.NotEmptyMax>
300003

EDC Ring Buffer Control Buffer Not Empty Max Accum

MOVE
Source RB.NotEmptyTMR.ACC
<EDC.RB_Cycle01.NotEmptyTMR.ACC>
0
Dest RB.NotEmptyMax
<EDC.RB_Cycle01.NotEmptyMax>
300003

Process RingBuffer Reads from Transaction Manager

RingBuffer Alarm

Alarm if the RingBuffer is not Empty for 5 minutes.
(This will provide time to resolve the problem BEFORE the RingBuffer is full and data is lost.)

21

EDC Ring Buffer Control Buffer Full
RB.Full
<EDC.RB_Cycle01.Full>

EDC Ring Buffer Control Buffer Not Empty Alarm Timer
RB.NotEmptyTMR.DN
<EDC.RB_Cycle01.NotEmptyTMR.DN>

EDC Ring Buffer Control Read Buffer Alarm
RB.BufferAlarm
<EDC.RB_Cycle01.BufferAlarm>

Copy Equipment working tags to Arrayed Tags
Working Tag is used for speed and logic readabliity (ie. can be xref'd)
DN007 Split EquipStatus Tag

22

EDC Ring Buffer Control Read Data has Valid Equipment ID
RB.ReadValidEQID
<EDC.RB_Cycle01.ReadValidEQID>

Server Interface Status Lines and Equipment Array

COP	
Source	EquipStatus01
Dest	EDC_Status01.Line[PerfLNi].Equip[PerfEQi]
Length	1

Tool, Trend, Summary Rx/Tx Status Lines and Equipment Array

COP	
Source	EquipStatus04
Dest	EDC_Status04.Line[PerfLNi].Equip[PerfEQi]
Length	1

(End)

RB Write -- Periodic Resend Transaction
Resend last message if RingBuffer is EMPTY for Keep Alive to TM/SQL
(Server will discard duplicate Transaction Numbers)

0

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE	
Source	1121
Dest	Prog.Routine 1120

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD	
Source A	Prog.RoutineScan[Prog.Routine] -806558467
Source B	1
Dest	Prog.RoutineScan[Prog.Routine] -806558467

Use Read Tag Pointer to store Read Record in Temp
(Read and Write records may be in different Tags at Rollovers)
Tags 01-05

2

EDC Ring Buffer Control Current/Last Read Tag

EQ	
Source A	RB.ReadTagPtr <EDC.RB_Cycle01.ReadTagPtr> 2
Source B	1

Equipment Data Cycle Message

COP	
Source	EDC_RB_Cycle01_Rec01[RB.ReadRecPtr]
Dest	KeepAliveMsg
Length	1

EDC Ring Buffer Control Current/Last Read Tag

EQ	
Source A	RB.ReadTagPtr <EDC.RB_Cycle01.ReadTagPtr> 2
Source B	2

Equipment Data Cycle Message

COP	
Source	EDC_RB_Cycle01_Rec02[RB.ReadRecPtr]
Dest	KeepAliveMsg
Length	1

EDC Ring Buffer Control Current/Last Read Tag

EQ	
Source A	RB.ReadTagPtr <EDC.RB_Cycle01.ReadTagPtr> 2
Source B	3

Equipment Data Cycle Message

COP	
Source	EDC_RB_Cycle01_Rec03[RB.ReadRecPtr]
Dest	KeepAliveMsg
Length	1

EDC Ring Buffer Control Current/Last Read Tag

EQ	
Source A	RB.ReadTagPtr <EDC.RB_Cycle01.ReadTagPtr> 2
Source B	4

Equipment Data Cycle Message

COP	
Source	EDC_RB_Cycle01_Rec04[RB.ReadRecPtr]
Dest	KeepAliveMsg
Length	1

EDC Ring Buffer Control Current/Last Read Tag

EQ	
Source A	RB.ReadTagPtr <EDC.RB_Cycle01.ReadTagPtr> 2
Source B	5

Equipment Data Cycle Message

COP	
Source	EDC_RB_Cycle01_Rec05[RB.ReadRecPtr]
Dest	KeepAliveMsg
Length	1

3

Increment Command Number to indicate Keep Alive Count
(used by TM Stored Procedure to detect difference between Keep Alive and Duplicate)

DHF024 Mar05, 2019 - Keep Alive CmdNum limited to 1-9999

Equipment Data
Cycle Message Header
- Command Number

ADD	
Source A	KeepAliveMsg.Header.CmdNum 1
Source B	1
Dest	KeepAliveMsg.Header.CmdNum 1

Equipment Data
Cycle Message Header
- Command Number

LT	
Source A	KeepAliveMsg.Header.CmdNum 1
Source B	1

Equipment Data
Cycle Message Header
- Command Number

GT	
Source A	KeepAliveMsg.Header.CmdNum 1
Source B	9999

Equipment Data
Cycle Message Header
- Command Number

MOVE	
Source	1
Dest	KeepAliveMsg.Header.CmdNum 1

4

Use Write Tag Pointer to store Temp Record
(Read and Write records may be in different Tags at Rollovers)
Tags 01-05

EDC Ring Buffer Control Next Write Tag

EQ
Source A RB.WriteTagPtr
<EDC.RB_Cycle01.WriteTagPtr> 2
Source B 1

Equipment Data Cycle Message

COP
Source KeepAliveMsg
Dest EDC_RB_Cycle01_Rec01[RB.WriteRecPtr]
Length 1

EDC Ring Buffer Control Next Write Tag

EQ
Source A RB.WriteTagPtr
<EDC.RB_Cycle01.WriteTagPtr> 2
Source B 2

Equipment Data Cycle Message

COP
Source KeepAliveMsg
Dest EDC_RB_Cycle01_Rec02[RB.WriteRecPtr]
Length 1

EDC Ring Buffer Control Next Write Tag

EQ
Source A RB.WriteTagPtr
<EDC.RB_Cycle01.WriteTagPtr> 2
Source B 3

Equipment Data Cycle Message

COP
Source KeepAliveMsg
Dest EDC_RB_Cycle01_Rec03[RB.WriteRecPtr]
Length 1

EDC Ring Buffer Control Next Write Tag

EQ
Source A RB.WriteTagPtr
<EDC.RB_Cycle01.WriteTagPtr> 2
Source B 4

Equipment Data Cycle Message

COP
Source KeepAliveMsg
Dest EDC_RB_Cycle01_Rec04[RB.WriteRecPtr]
Length 1

EDC Ring Buffer Control Next Write Tag

EQ
Source A RB.WriteTagPtr
<EDC.RB_Cycle01.WriteTagPtr> 2
Source B 5

Equipment Data Cycle Message

COP
Source KeepAliveMsg
Dest EDC_RB_Cycle01_Rec05[RB.WriteRecPtr]
Length 1

5

Log Transaction

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB.WriteTagPtr
<EDC.RB_Cycle01.WriteTagPtr>
2
Dest Trans.Opt[1]
<EDC.Trans.Opt[1]>
-1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB.WriteRecPtr
<EDC.RB_Cycle01.WriteRecPtr>
547
Dest Trans.Opt[2]
<EDC.Trans.Opt[2]>
-1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB.ReadTagPtr
<EDC.RB_Cycle01.ReadTagPtr>
2
Dest Trans.Opt[3]
<EDC.Trans.Opt[3]>
-1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB.ReadRecPtr
<EDC.RB_Cycle01.ReadRecPtr>
546
Dest Trans.Opt[4]
<EDC.Trans.Opt[4]>
-1

Log Transactions

JSR
Routine Name _901_TransLogSBR
Input Par 2522
Input Par 0
Input Par LNi
Input Par EQi
Input Par RB.TransactionNumber
<EDC.RB_Cycle01.TransactionNumber>

EDC Ring Buffer Control Read Data has Valid Equipment ID
RB.ReadValidEQID
<EDC.RB_Cycle01.ReadValidEQID>

Equipment Data Collection - Line Equipment Control for Message Transaction Timer Accum Last (used to calculate differential)

CLR
Dest TempPerfMsg.TransLast
110

Equipment Data Collection - Line Equipment Control for Message Transaction Timer

MOVE
Source RB.KeepAliveTMR.ACC
<EDC.RB_Cycle01.KeepAliveTMR.ACC>
34823
Dest TempPerfMsg.TransTMR.ACC
110

Performance Transaction

aoiEDC_PerfTrans
aoiEDC_PerfTrans EDC_aoiPerfTrans ...
PerfMsg TempPerfMsg
PerfTrans EquipStatus04.CycleTx.KA

Increment Write Pointer

6

EDC Ring Buffer Control Next Write Record

ADD
Source A RB.WriteRecPtr <EDC.RB_Cycle01.WriteRecPtr> 547
Source B 1
Dest RB.WriteRecPtr <EDC.RB_Cycle01.WriteRecPtr> 547

EDC Ring Buffer Control Next Write Record

LT
Source A RB.WriteRecPtr <EDC.RB_Cycle01.WriteRecPtr> 547
Source B 0

EDC Ring Buffer Control Next Write Record

GT
Source A RB.WriteRecPtr <EDC.RB_Cycle01.WriteRecPtr> 547
Source B RB.MaxRecPtr <EDC.RB_Cycle01.MaxRecPtr> 999

EDC Ring Buffer Control Next Write Record

MOVE
Source 0
Dest RB.WriteRecPtr <EDC.RB_Cycle01.WriteRecPtr> 547

EDC Ring Buffer Control Next Write Tag

ADD
Source A RB.WriteTagPtr <EDC.RB_Cycle01.WriteTagPtr> 2
Source B 1
Dest RB.WriteTagPtr <EDC.RB_Cycle01.WriteTagPtr> 2

EDC Ring Buffer Control Next Write Tag

LT
Source A RB.WriteTagPtr <EDC.RB_Cycle01.WriteTagPtr> 2
Source B 1

EDC Ring Buffer Control Next Write Tag

GT
Source A RB.WriteTagPtr <EDC.RB_Cycle01.WriteTagPtr> 2
Source B RB.MaxTagPtr <EDC.RB_Cycle01.MaxTagPtr> 5

EDC Ring Buffer Control Next Write Tag

MOVE
Source 1
Dest RB.WriteTagPtr <EDC.RB_Cycle01.WriteTagPtr> 2

(End)

Log Transactions

Transaction Log

Capture Event for each Transaction
Code = Transaction Code (see Excel worksheet for definitions)
Time = Transaction Time (msec)
PNTi = Transaction Point Index (-1 = n/a)

0

Transaction Log Control SBR Input Parameter -- Transaction Code

SBR
Input Par Trans.Code
<EDC.Trans.Code>
Input Par Trans.Time
<EDC.Trans.Time>
Input Par Trans.LNi
<EDC.Trans.LNi>
Input Par Trans.EQi
<EDC.Trans.EQi>
Input Par Trans.Num
<EDC.Trans.Num>

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

AFI Prog.Routine to prevent TransLogSBR overwrite of calling Routine - obscures casue of Program Faults

1

AFI

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 901
Dest Prog.Routine 1120

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[901] 91892768
Source B 1
Dest Prog.RoutineScan[901] 91892768

Log Transactions

Exit if Transaction Code, LNi and EQi are not within filtered logging values.

2

Transaction Log Control Enable Code Filters
Trans.Filter.CodeEN
<EDC.Trans.Filter.CodeEN>

Transaction Log Control Enable LNi Filters
Trans.Filter.LNiEN
<EDC.Trans.Filter.LNiEN>

Transaction Log Control Enable EQi Filters
Trans.Filter.EQiEN
<EDC.Trans.Filter.EQiEN>

Transaction Log Control Log Record
Trans.Filter.Log
<EDC.Trans.Filter.Log>

Transaction Log Control SBR Input Parameter -- Transaction Code

LIMIT
Low Limit Trans.Filter.CodeLow
<EDC.Trans.Filter.CodeLow>
0
Test Trans.Code
<EDC.Trans.Code>
2021
High Limit Trans.Filter.CodeHigh
<EDC.Trans.Filter.CodeHigh>
0

Transaction Log Control SBR Input Parameter -- Line Index

LIMIT
Low Limit Trans.Filter.LNiLow
<EDC.Trans.Filter.LNiLow>
0
Test Trans.LNi
<EDC.Trans.LNi>
4
High Limit Trans.Filter.LNiHigh
<EDC.Trans.Filter.LNiHigh>
0

Transaction Log Control SBR Input Parameter -- Equipment Index

LIMIT
Low Limit Trans.Filter.EQiLow
<EDC.Trans.Filter.EQiLow>
0
Test Trans.EQi
<EDC.Trans.EQi>
21
High Limit Trans.Filter.EQiHigh
<EDC.Trans.Filter.EQiHigh>
0

Transaction Log Control Log Record
Trans.Filter.Log
<EDC.Trans.Filter.Log>

lblExit
JMP

Log Transactions

Increment Log Index and check for legal value

3

Transaction Log Control -- Transaction Log Index

ADD
Source A Trans.LOGi <EDC.Trans.LOGi> 211
Source B 1
Dest Trans.LOGi <EDC.Trans.LOGi> 211

Transaction Log Control -- Transaction Log Index

LT
Source A Trans.LOGi <EDC.Trans.LOGi> 211
Source B 0

Transaction Log Control -- Transaction Log Index

GE
Source A Trans.LOGi <EDC.Trans.LOGi> 211
Source B CFG.TransLOG.n <EDC.CFG.TransLOG.n> 250

Transaction Log Control -- Transaction Log Index

CLR
Dest Trans.LOGi <EDC.Trans.LOGi> 211

Generate Log Index for clearing Next Transaction

4

Transaction Log Control -- Transaction Log Index +1 (for clearing next record)

ADD
Source A Trans.LOGi <EDC.Trans.LOGi> 211
Source B 1
Dest Trans.LOGx <EDC.Trans.LOGx> 212

Transaction Log Control -- Transaction Log Index +1 (for clearing next record)

LT
Source A Trans.LOGx <EDC.Trans.LOGx> 212
Source B 0

Transaction Log Control -- Transaction Log Index +1 (for clearing next record)

GE
Source A Trans.LOGx <EDC.Trans.LOGx> 212
Source B CFG.TransLOG.n <EDC.CFG.TransLOG.n> 250

Transaction Log Control -- Transaction Log Index +1 (for clearing next record)

CLR
Dest Trans.LOGx <EDC.Trans.LOGx> 212

Log Transactions

Initialize Transaction and Next Transaction

5

Transaction Log

FLL	
Source	0
Dest	EDC_TransLog[Trans.LOGi]
Length	1

Transaction Log

FLL	
Source	0
Dest	EDC_TransLog[Trans.LOGx]
Length	1

Store Transaction Header
-- Transaction Code, Time, Point Index
-- DateStamp (YYMMDD), TimeStamp (HHMMSS), TimeMicroSec (microseconds), Scan Counter

6

Transaction Log

MOVE	
Source	Trans.Code
	<EDC.Trans.Code>
	2021
Dest	EDC_TransLog[Trans.LOGi].TransCode
	1011

Transaction Log

MOVE	
Source	Trans.Time
	<EDC.Trans.Time>
	0
Dest	EDC_TransLog[Trans.LOGi].TransTime
	0

Transaction Log

MOVE	
Source	Trans.LNi
	<EDC.Trans.LNi>
	4
Dest	EDC_TransLog[Trans.LOGi].LNi
	15

Transaction Log

MOVE	
Source	Trans.EQi
	<EDC.Trans.EQi>
	21
Dest	EDC_TransLog[Trans.LOGi].EQi
	17

Transaction Log

MOVE	
Source	Trans.Num
	<EDC.Trans.Num>
	4971
Dest	EDC_TransLog[Trans.LOGi].Num
	5631

Transaction Log

MOVE	
Source	CLX.SysTime[8]
	<EDC.CLX.SysTime[8]>
	20261030
Dest	EDC_TransLog[Trans.LOGi].DateStamp
	20261030

Transaction Log

MOVE	
Source	CLX.SysTime[9]
	<EDC.CLX.SysTime[9]>
	151921
Dest	EDC_TransLog[Trans.LOGi].TimeStamp
	152006

Transaction Log

MOVE	
Source	CLX.SysTime[6]
	<EDC.CLX.SysTime[6]>
	904687
Dest	EDC_TransLog[Trans.LOGi].TimeMicroSec
	955653

Transaction Log

MOVE	
Source	CLX.ScanCount
	<EDC.CLX.ScanCount>
	466525176
Dest	EDC_TransLog[Trans.LOGi].ScanCount
	466528353

Store Optional Transaction Details
(loaded before calling Log Routine)

7

Transaction Log

MOVE	
Source	Trans.Opt[1]
	<EDC.Trans.Opt[1]>
	-1
Dest	EDC_TransLog[Trans.LOGi].Opt1
	1517

Transaction Log

MOVE	
Source	Trans.Opt[2]
	<EDC.Trans.Opt[2]>
	-1
Dest	EDC_TransLog[Trans.LOGi].Opt2
	0

Transaction Log

MOVE	
Source	Trans.Opt[3]
	<EDC.Trans.Opt[3]>
	-1
Dest	EDC_TransLog[Trans.LOGi].Opt3
	-1

Transaction Log

MOVE	
Source	Trans.Opt[4]
	<EDC.Trans.Opt[4]>
	-1
Dest	EDC_TransLog[Trans.LOGi].Opt4
	-1

Transaction Log

MOVE	
Source	Trans.Opt[5]
	<EDC.Trans.Opt[5]>
	-1
Dest	EDC_TransLog[Trans.LOGi].Opt5
	-1

Log Transactions

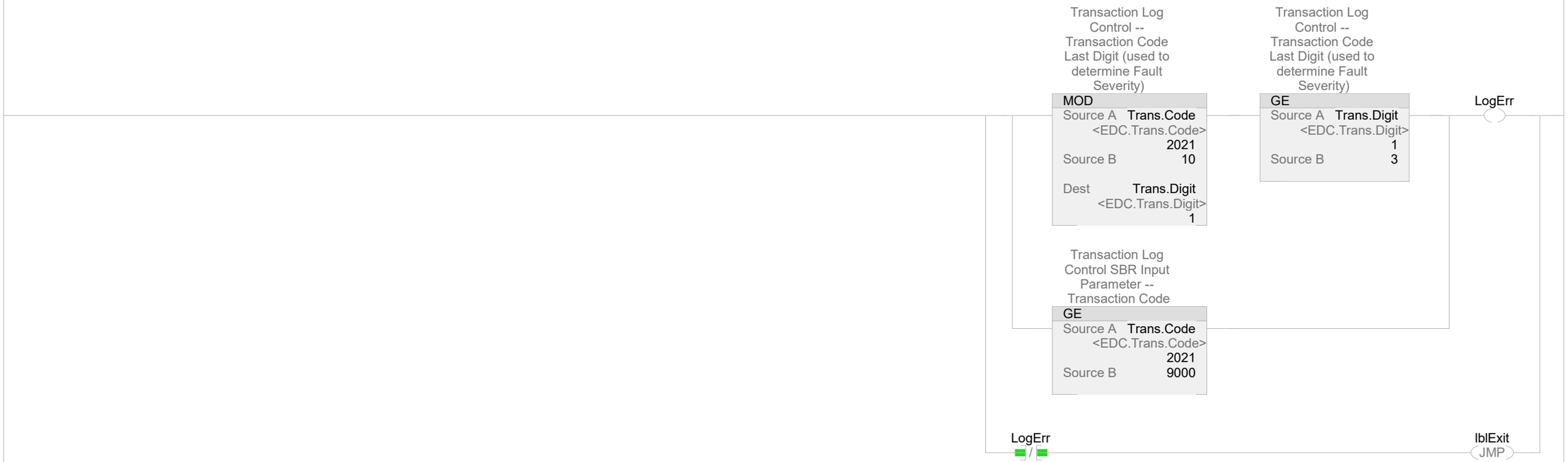
Error Transaction Log

Capture Event for each Error Transaction in separate Error Transaction Log
(preserves errors from being overwritten by normal transactions)

Events codes should be assigned as follows:
xxx0 = Enable -- Yellow Status for Normal Operation in Process
xxx1 = Done -- Green Status for Normal Operation Completed
xxx2 = Retry -- Orange Status for Normal Operation in Retry
xxx3 = Error -- Red Status for Normal Operation Failed
xxx4-xxx9 = Other Errors -- Read Status (see TransLog Definitions)

9xxx = Initialization Events

8



Error Transaction Log

Capture Event for each Error Transaction in separate Error Transaction Log
(preserves errors from being overwritten by normal transactions)

9

NOP

Log Transactions

Increment Error Log Index and check for legal value

10

Transaction Log Control -- Transaction Error Log Index

ADD
Source A Trans.ERRi <EDC.Trans.ERRi> 62
Source B 1
Dest Trans.ERRi <EDC.Trans.ERRi> 62

Transaction Log Control -- Transaction Error Log Index

LT
Source A Trans.ERRi <EDC.Trans.ERRi> 62
Source B 0

Transaction Log Control -- Transaction Error Log Index

GE
Source A Trans.ERRi <EDC.Trans.ERRi> 62
Source B CFG.TransERR.n <EDC.CFG.TransERR.n> 100

Transaction Log Control -- Transaction Error Log Index

CLR
Dest Trans.ERRi <EDC.Trans.ERRi> 62

Generate Error Log Index for clearing Next Transaction

11

Transaction Log Control -- Transaction Error Log Index +1 (for clearing next record)

ADD
Source A Trans.ERRi <EDC.Trans.ERRi> 62
Source B 1
Dest Trans.ERRx <EDC.Trans.ERRx> 63

Transaction Log Control -- Transaction Error Log Index +1 (for clearing next record)

LT
Source A Trans.ERRx <EDC.Trans.ERRx> 63
Source B 0

Transaction Log Control -- Transaction Error Log Index +1 (for clearing next record)

GE
Source A Trans.ERRx <EDC.Trans.ERRx> 63
Source B CFG.TransERR.n <EDC.CFG.TransERR.n> 100

Transaction Log Control -- Transaction Error Log Index +1 (for clearing next record)

CLR
Dest Trans.ERRx <EDC.Trans.ERRx> 63

Log Transactions

Copy Transaction to Transaction Error Log and Initialize Next Transaction Error Log

12

Error Log

COP
Source EDC_TransLog[Trans.LOGi]
Dest EDC_TransErr[Trans.ERRi]
Length 1

Error Log

FLL
Source 0
Dest EDC_TransErr[Trans.ERRx]
Length 1

Exit Routine

13

lblExit
LBL

NOP

Clear Optional Transaction Details
Clear i to avoid bad indexes on next Transaction

14

Transaction Log Control SBR External Load -- Optional Parameters

FLL
Source -1
Dest Trans.Opt[0]
<EDC.Trans.Opt[0]>
Length Trans.OptN
<EDC.Trans.OptN>

(End)

eAssy_EDC_Gway
MainTask
RingBufferCycle01
_1101_Main
Ladder Diagram 1
_1102_Fault
Ladder Diagram 4
_1111_Config
Ladder Diagram 8
_1120_RingBuffer
Ladder Diagram 10
_1121_RingBufferKeepAlive
Ladder Diagram 29
_901_TransLogSBR
Ladder Diagram 35